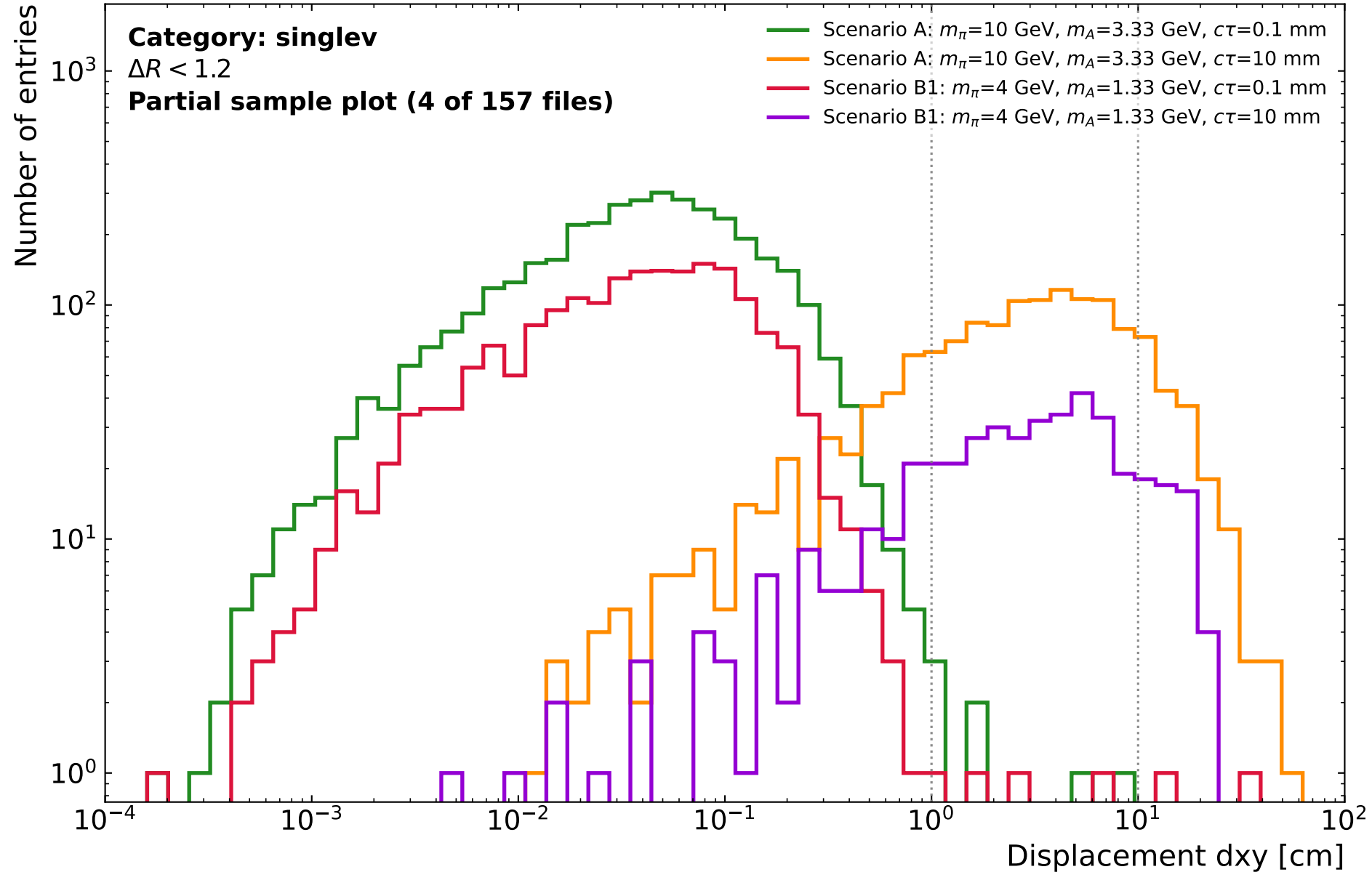


singlev: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary**

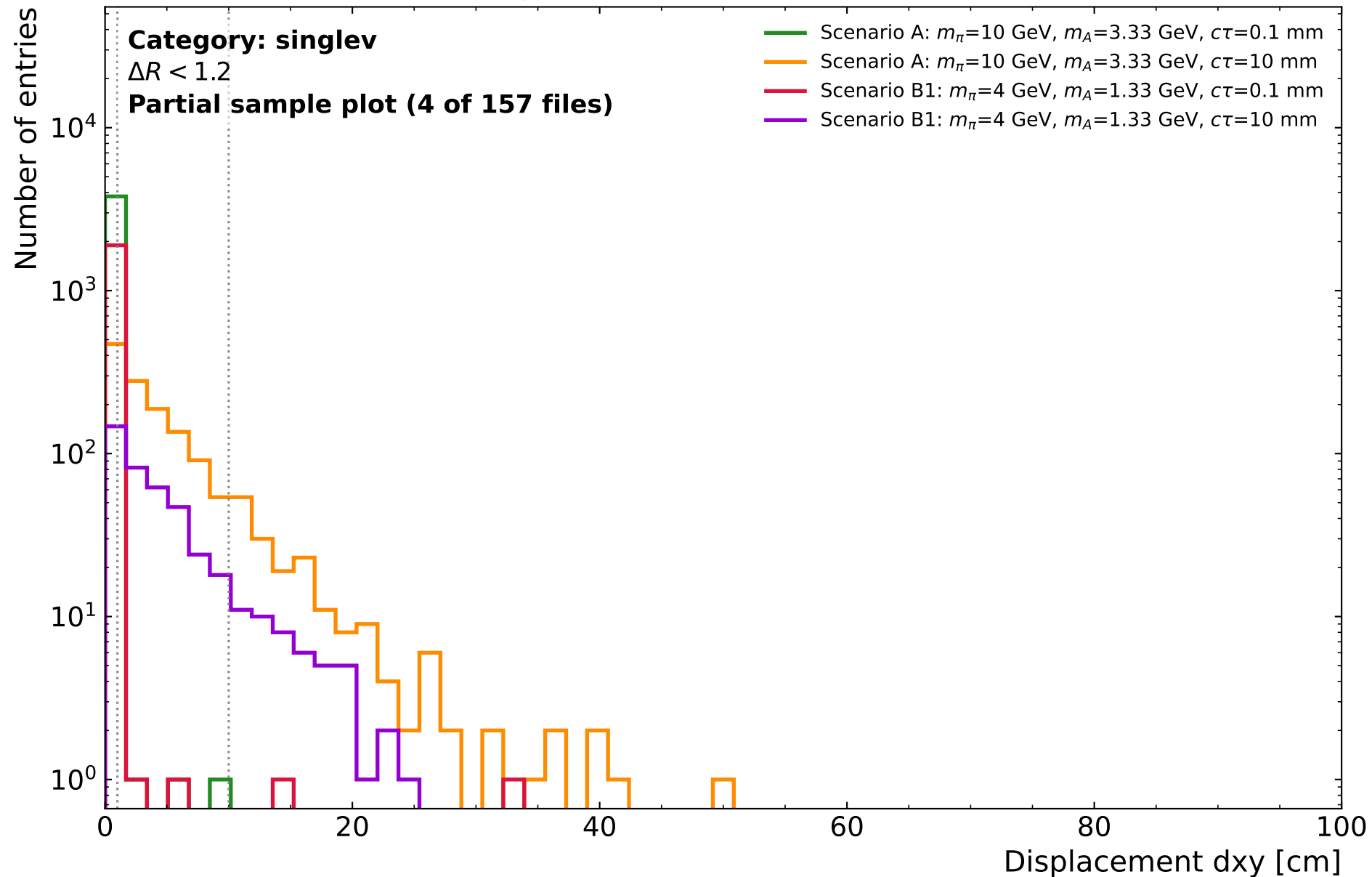
(13.6 TeV, 2024)



singlev: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

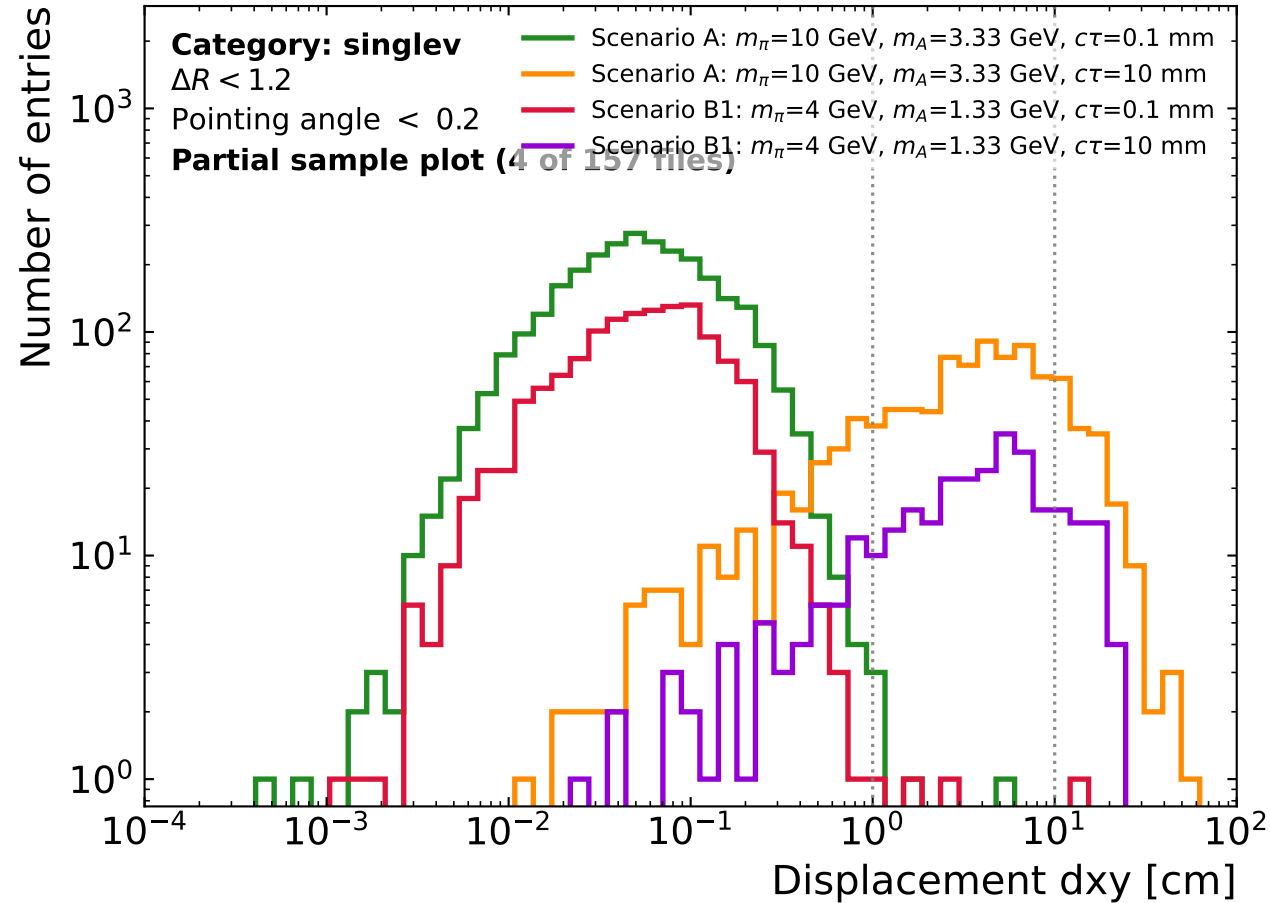
**CMS** *Simulation Preliminary*

(13.6 TeV, 2024)



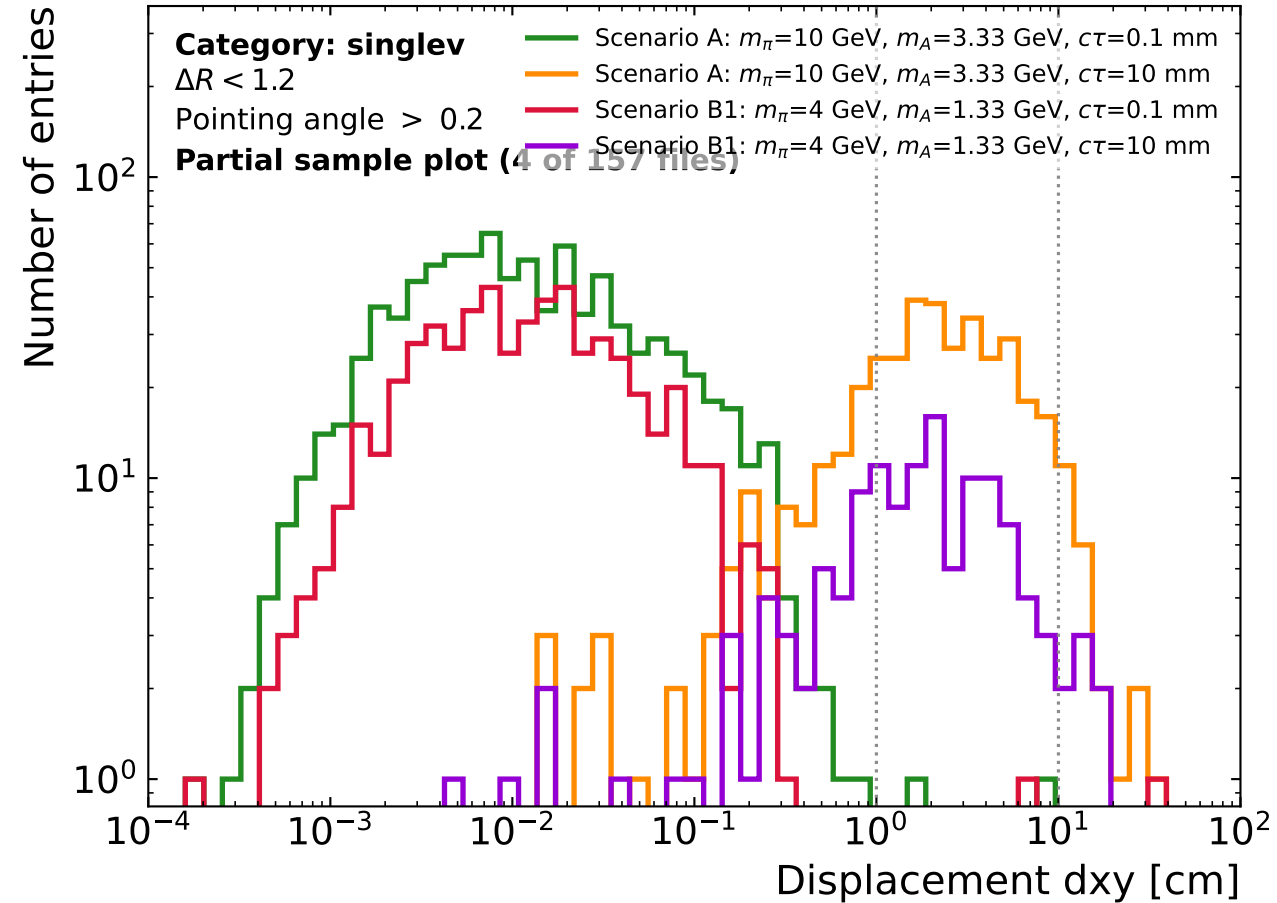
singlev: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index), split by pointing angle  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary** (13.6 TeV, 2024)



|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 2,883   | 3     | 0        | 2,886 |
| scA $c\tau=10$   | 212     | 637   | 157      | 1,006 |
| scB1 $c\tau=0.1$ | 1,350   | 2     | 1        | 1,353 |
| scB1 $c\tau=10$  | 52      | 202   | 45       | 299   |

**CMS Simulation Preliminary** (13.6 TeV, 2024)

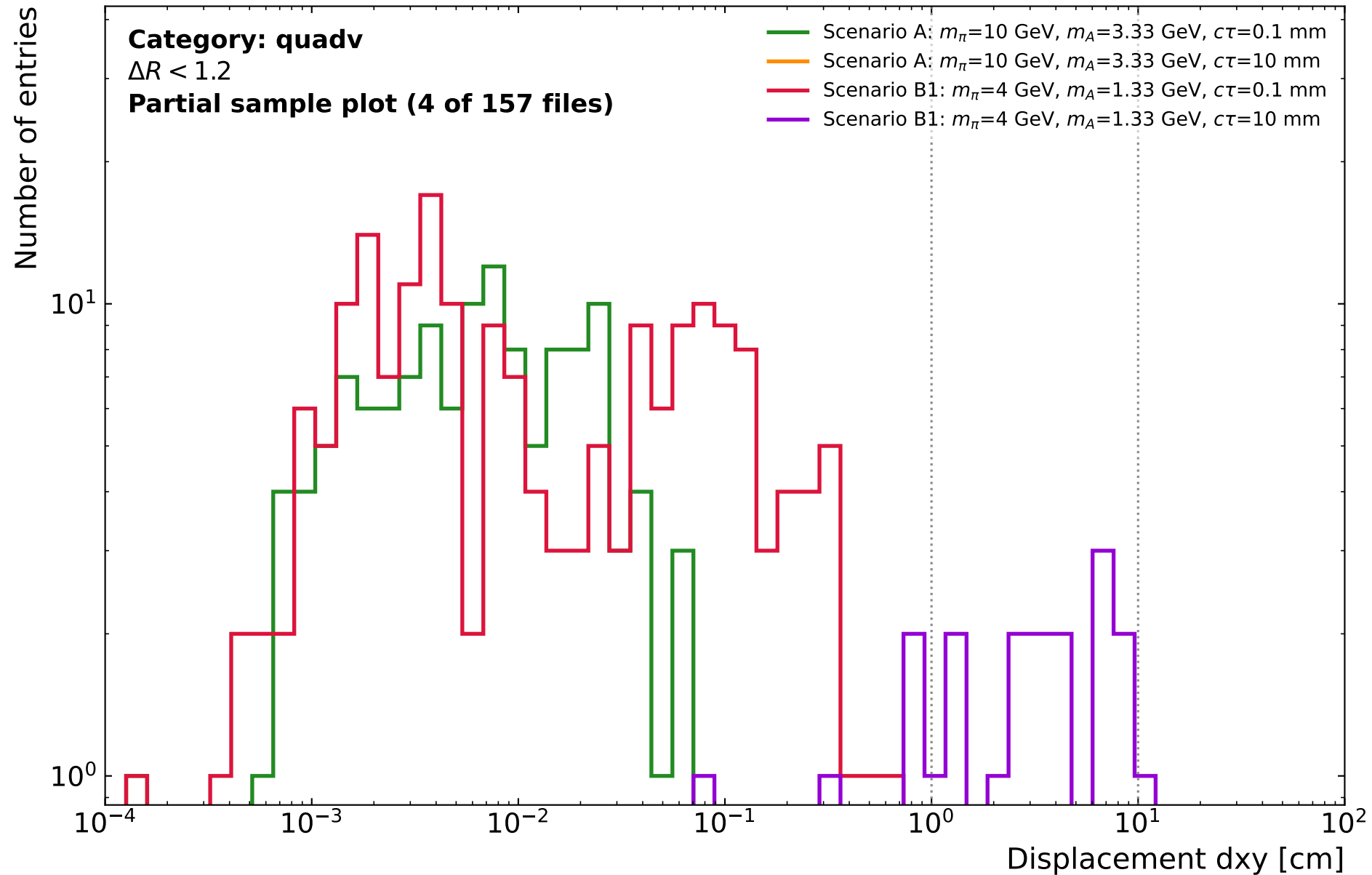


|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 901     | 2     | 0        | 903   |
| scA $c\tau=10$   | 102     | 265   | 23       | 390   |
| scB1 $c\tau=0.1$ | 547     | 1     | 1        | 549   |
| scB1 $c\tau=10$  | 41      | 82    | 7        | 130   |

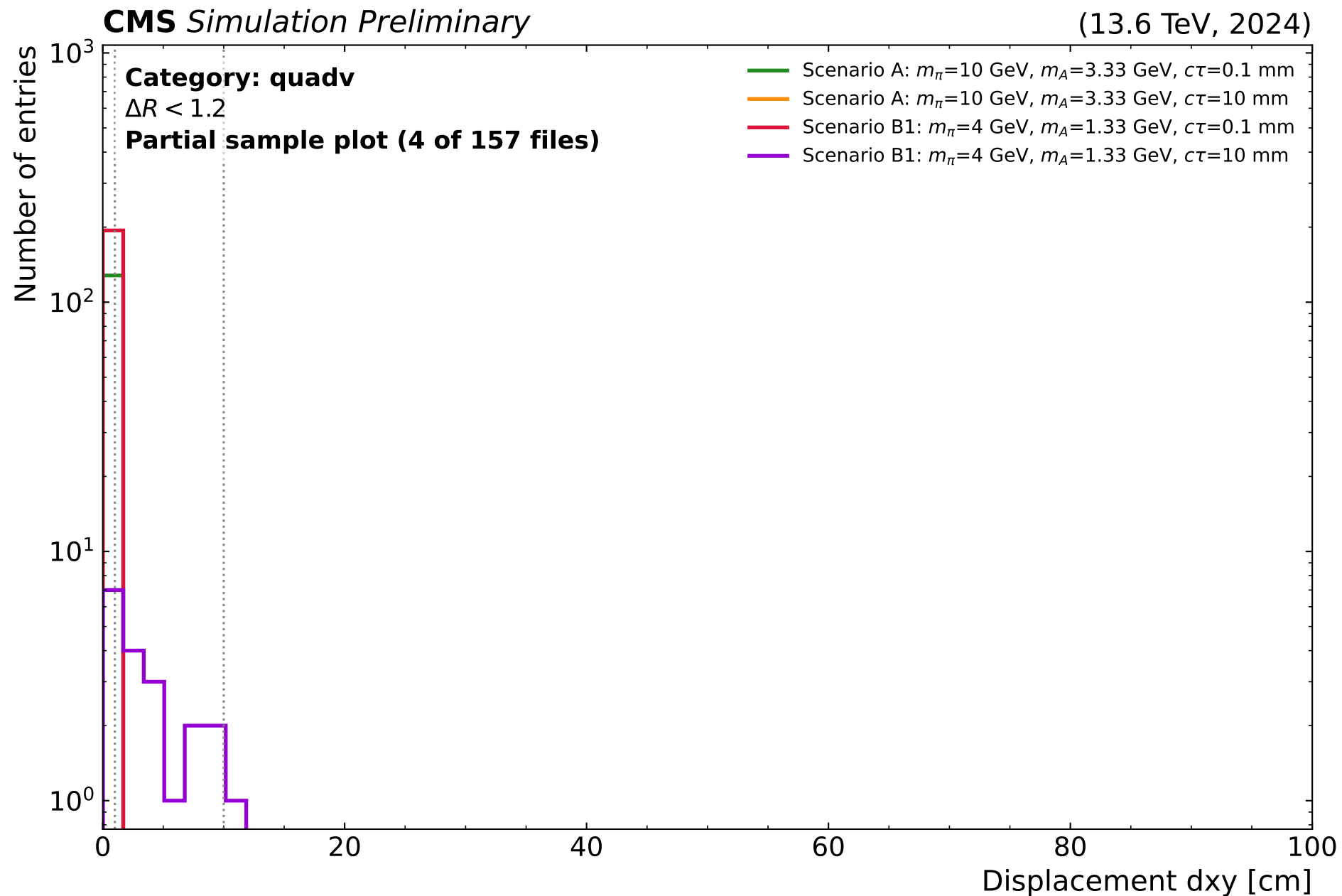
quadv: displacement of the binning vertex (fourmuonSV\_dxy)  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary**

(13.6 TeV, 2024)

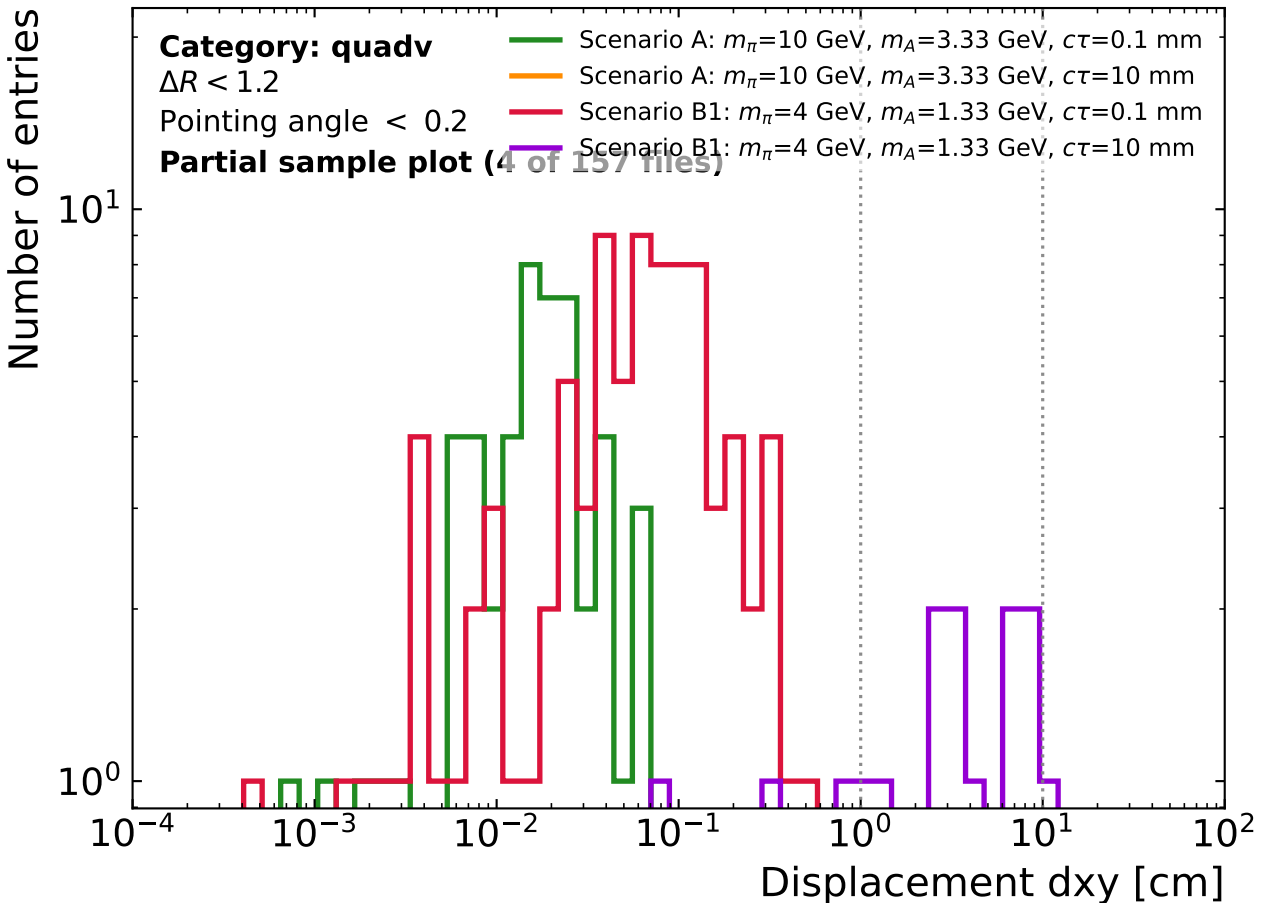


quadv: displacement of the binning vertex (fourmuonSV\_dxy)  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)



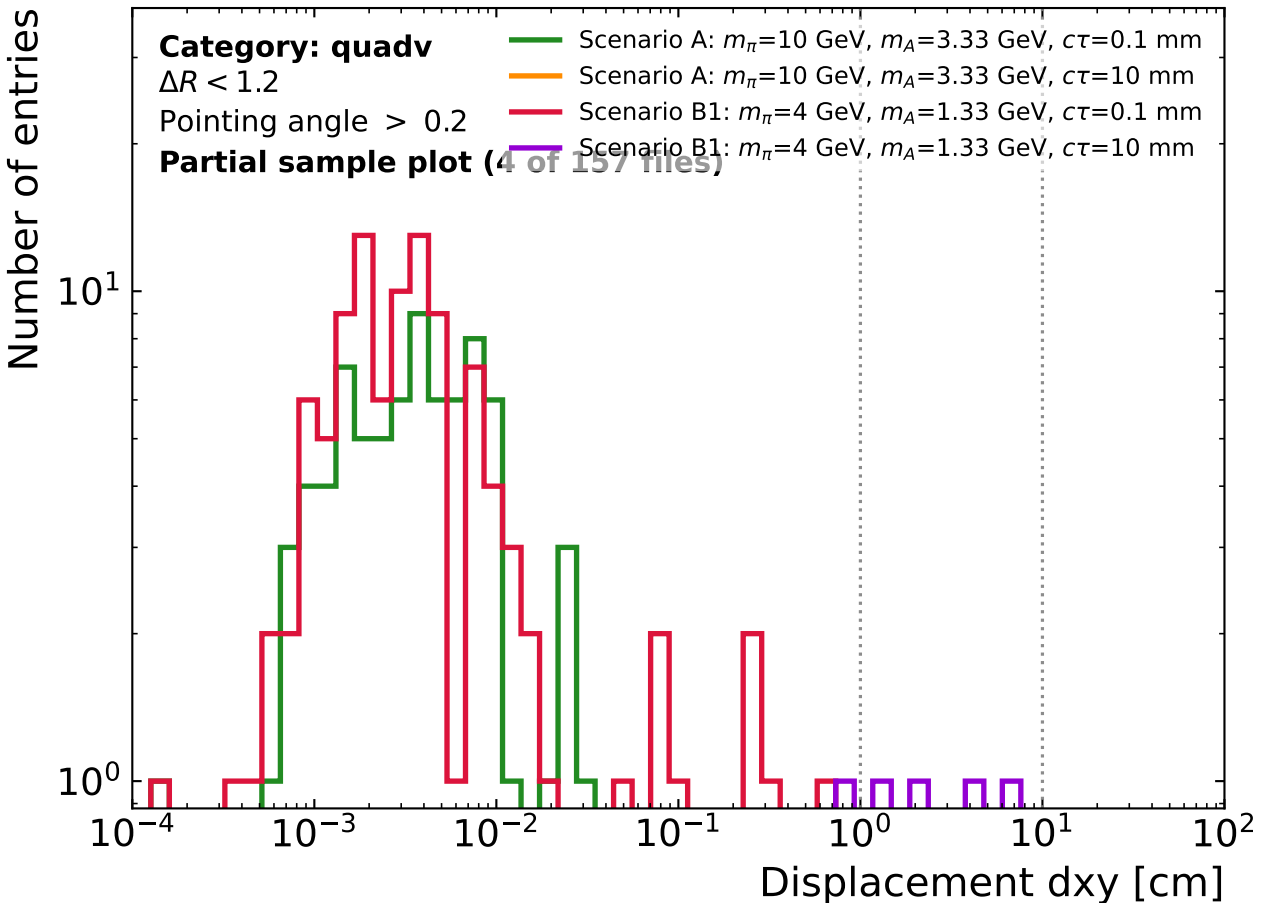
quadv: displacement of the binning vertex (fourmuonSV\_dxy), split by pointing angle  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary** (13.6 TeV, 2024)



|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 51      | 0     | 0        | 51    |
| scA $c\tau=10$   | 0       | 0     | 0        | 0     |
| scB1 $c\tau=0.1$ | 90      | 0     | 0        | 90    |
| scB1 $c\tau=10$  | 3       | 11    | 1        | 15    |

**CMS Simulation Preliminary** (13.6 TeV, 2024)

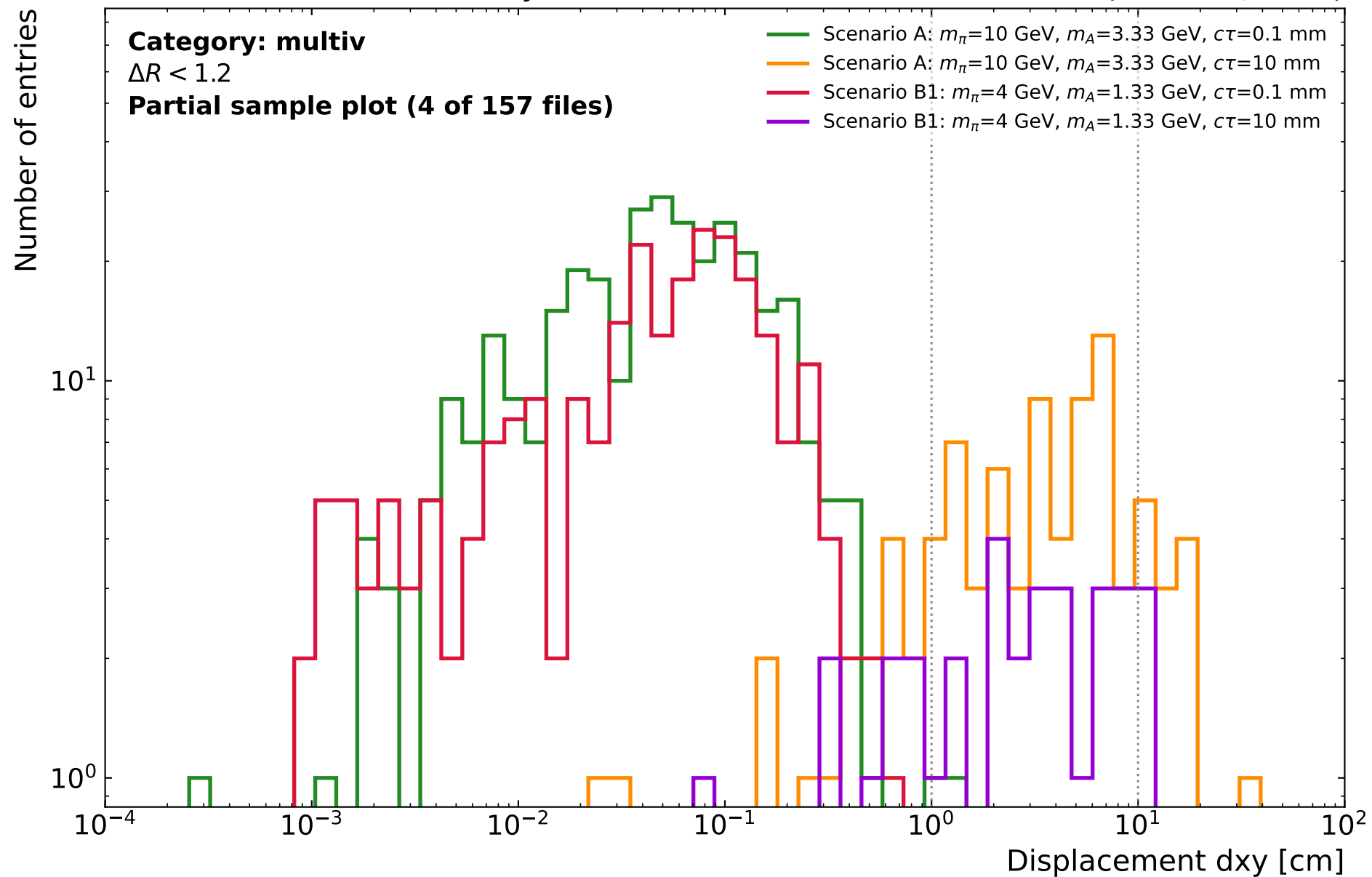


|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 77      | 0     | 0        | 77    |
| scA $c\tau=10$   | 0       | 0     | 0        | 0     |
| scB1 $c\tau=0.1$ | 104     | 0     | 0        | 104   |
| scB1 $c\tau=10$  | 1       | 4     | 0        | 5     |

multiv: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

**CMS** *Simulation Preliminary*

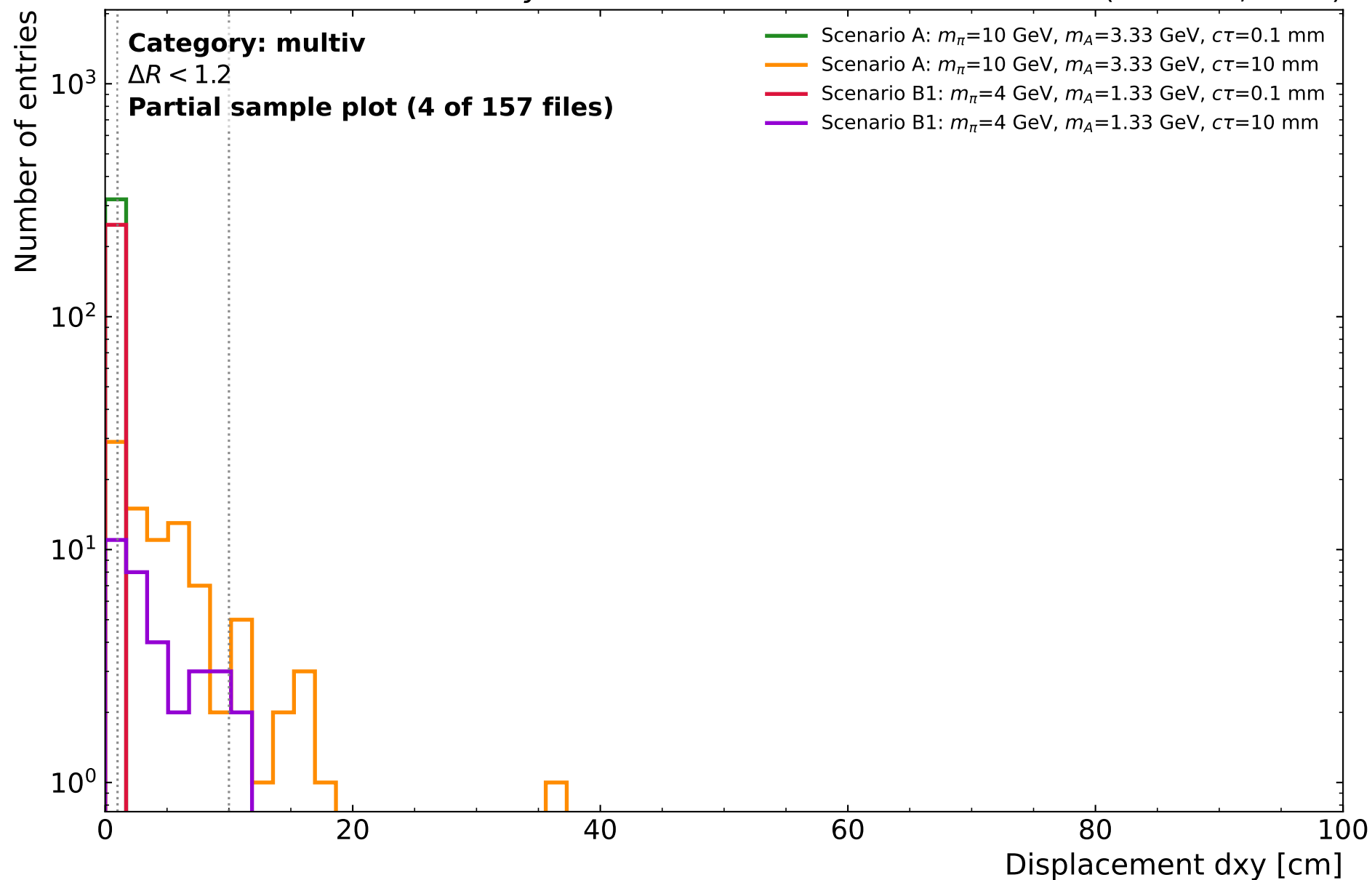
(13.6 TeV, 2024)



multiv: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)

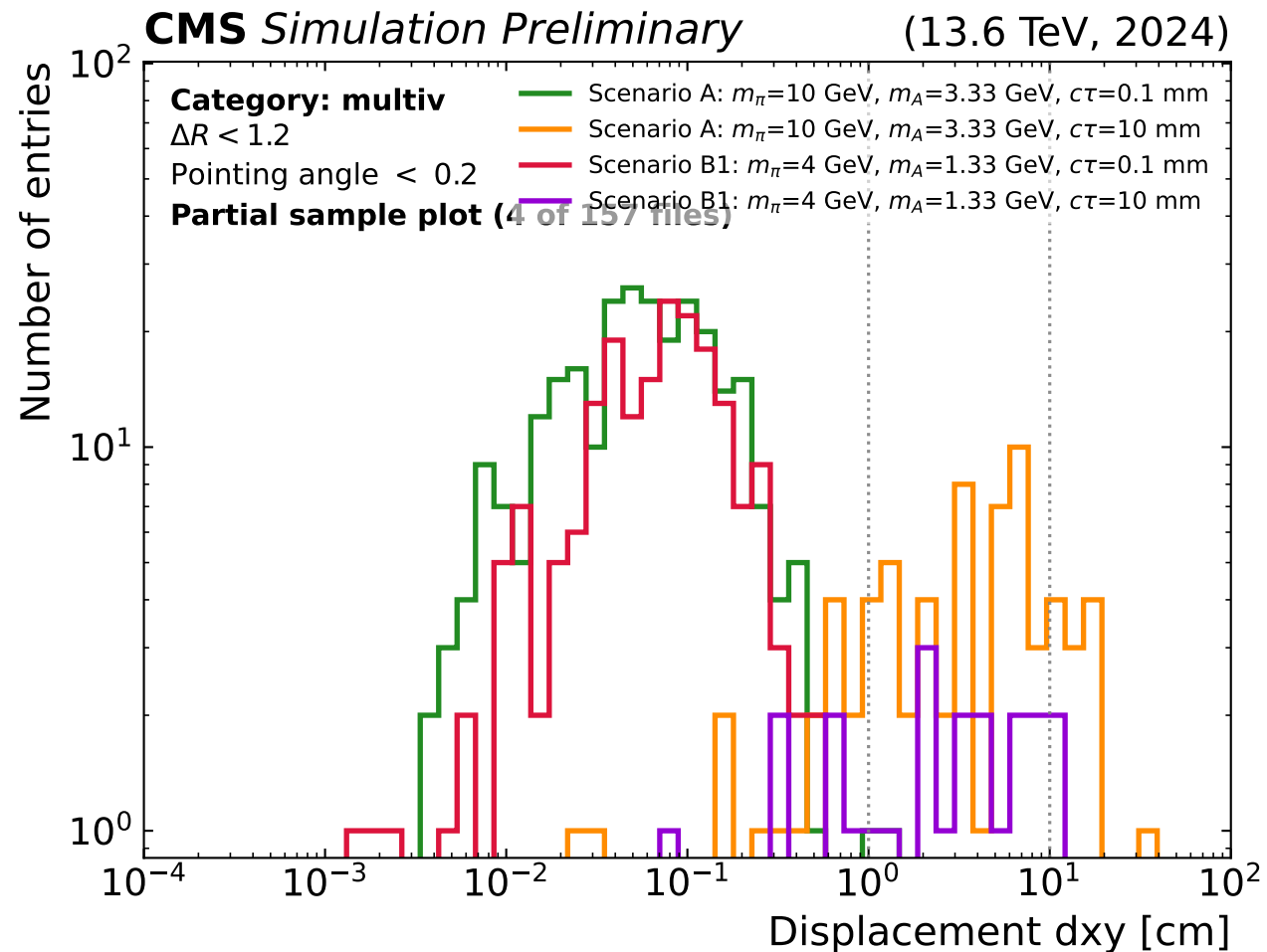
**CMS** *Simulation Preliminary*

(13.6 TeV, 2024)

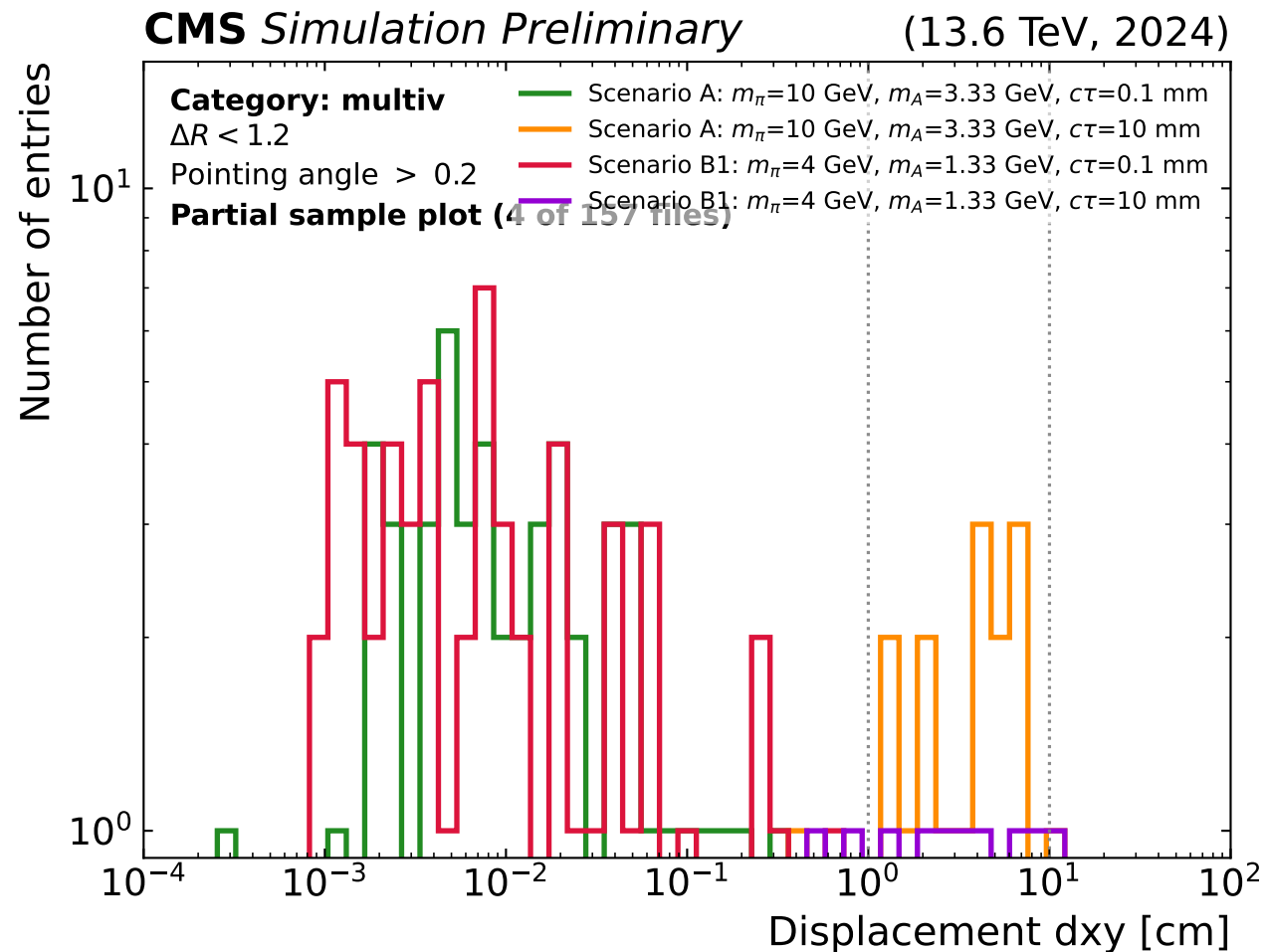




multiv: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index), split by pointing angle  
 $\Delta R < 1.2$ ; dotted: dxy category edges (1, 10 cm)



|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 266     | 2     | 0        | 268   |
| scA $c\tau=10$   | 17      | 44    | 12       | 73    |
| scB1 $c\tau=0.1$ | 190     | 0     | 0        | 190   |
| scB1 $c\tau=10$  | 6       | 15    | 2        | 23    |



|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 51      | 0     | 0        | 51    |
| scA $c\tau=10$   | 1       | 15    | 1        | 17    |
| scB1 $c\tau=0.1$ | 58      | 0     | 0        | 58    |
| scB1 $c\tau=10$  | 2       | 7     | 1        | 10    |